

Spatiotemporal Analysis and Severity Prediction of Traffic Collisions in Los Angeles

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Section 1: Project Definition

Los Angeles is highly automobile-dependent, with a large road network, heavy traffic, and long commute distances. However, traffic safety analyses often focus only on total collision counts, which can overlook how severity varies by time, location, victim demographics, and collision conditions.

This project examines whether a multi-class classification model can predict traffic collision severity for specific demographic groups, such as young female victims, using temporal, spatial, demographic, and collision-condition features. Rather than predicting whether a collision will occur, the project focuses on how severe a collision is likely to be, with particular attention to fatal and severe injury cases. The goal is to identify high-risk periods, locations, victim groups, and conditions to better support traffic safety analysis in Los Angeles.

Section 2: Descriptions

2.1 Description of Background

Traffic collisions are not randomly distributed. Previous studies show that crashes often cluster in certain locations and time periods (Alam & Tabassum, 2023; Alsahfi, 2024). Research on crash hotspots, GIS-based traffic analysis, and severity-based spatial modeling supports the use of spatial and temporal features to understand traffic risk. Building on this research, our project combines spatial analysis, temporal feature engineering, demographic information, and machine learning to predict collision severity. The project focuses on high-severity outcomes rather than only total collision counts, because areas with the most crashes are not always the areas with the highest fatal or severe injury risk.

2.2 Description of the Dataset

This project uses Traffic Collision Data from 2010 to the present, published by the Los Angeles Police Department through the LA Open Data Portal. The raw dataset contains 621,677 records of traffic collisions in Los Angeles.

The dataset includes collision date and time, geographic coordinates, LAPD area division, victim age, victim sex, victim descent, MO Codes, and collision severity. From these fields, we created time-based features such as Hour, TimePeriod, and IsWeekend, as well as

demographic groupings and 11 MO Code indicator variables. Collision severity was encoded into five levels: Fatal, Severe Injury, Visible Injury, Complaint of Pain, and No Injury.

After cleaning invalid coordinates, missing MO Codes, missing age, descent, and sex, duplicate records, out-of-boundary locations, and remaining null values, 169,031 records were removed, leaving 452,646 records for analysis.

2.3 Description of Methods Used

The project uses three main methods: data preprocessing, spatial analysis, and predictive modeling. First, the raw data was cleaned and transformed into model-ready features. This included creating temporal variables, demographic groups, MO Code indicators, and a five-level severity target. Second, spatial analysis was conducted using Kernel Density Estimation and DBSCAN. Kernel Density Estimation was used to identify both general collision hotspots and fatal or severe injury hotspots. DBSCAN was used to remove spatial noise and identify meaningful collision clusters. These results were used to create a Spatial Risk Score.

Finally, a Random Forest multi-class classification model was trained to predict collision severity. The model was evaluated using accuracy, precision, recall, and a confusion matrix, with special attention to fatal and severe injury cases.

Section 3: Experiment Setup and Analysis Results

3.1 Experiment Setup

The experiment was designed to predict five levels of traffic collision severity in Los Angeles, which were extracted from LAPD MO Codes, using a priority order in cases where multiple severity codes appeared in the same record: 3027 (Fatal), 3024 (Severe Injury), 3025 (Visible Injury), 3026 (Complaint of Pain), and 3028 (Non-Injury). The final labeled dataset contained 360,687 records (“Unknown” not counted), including 3,059 Fatal cases, 14,797 Severe Injury cases, 58,646 Visible Injury cases, 135,877 Complaint of Pain cases, and 148,308 Non-Injury cases. This distribution shows a strong class imbalance, with Fatal collisions representing only about 0.85% of the labeled dataset, while Non-Injury accounts for 43.3%.

The experiment used 17 input variables across four categories: temporal features, demographic features, spatial risk, and collision type indicators. Temporal features included Hour, DayOfWeek, IsWeekend, Month, and TimePeriod. Demographic features included AgeGroup, Sex, and Descent. Collision-condition features were derived from MO Codes and included indicators for vehicle-versus-pedestrian, vehicle-versus-bike, vehicle-versus-vehicle,

vehicle-versus-motorcycle, hit-and-run, intersection-related collisions, street racing, and DUI. A spatial feature, `spatial_risk_score`, was added to represent area-level fatality risk.

The dataset was split into training, validation, and test sets using stratified sampling to preserve the severity distribution across all subsets. The training set contained 252,480 records or 70% of the data. The validation set contained 54,103 records, and the test set contained 54,104 records, each representing approximately 15% of the dataset. Categorical variables were label-encoded before model training.

Because the severity classes were highly imbalanced, the model was trained with `class_weight='balanced'`. Fatal collisions made up only about 0.85% of the dataset, while Non-Injury cases were the largest class. Without adjustment, the model could focus mostly on the majority class and miss rare but important high-severity cases. The balanced class weight gave higher importance to minority classes, with Fatal records receiving about 48 times more weight than Non-Injury records and Severe Injury records receiving about 10 times more weight.

The Random Forest classifier was trained using 200 decision trees, a maximum depth of 20, a minimum sample per leaf of 5, `class_weight='balanced'`, and `random_state=42` for reproducibility.

3.2 Temporal Analysis Results

Time analysis reveals that the periods with a high number of accidents and those with a high risk of death are not consistent. Most accidents occur between 2 p.m. and 6 p.m., but the fatality rate is the highest around 5 a.m. (about 2.55%, with an average of only 0.4% during the day). The overall fatality rate is the highest among all time periods in the late night (0-6 a.m.) (Figure 1). Further cross-analysis revealed that the fatality rate of the combination of late-night and vehicle-pedestrian collisions reached 13.14%, the highest among all combinations, indicating a significant interaction effect between time periods and collision types (Figure 2).

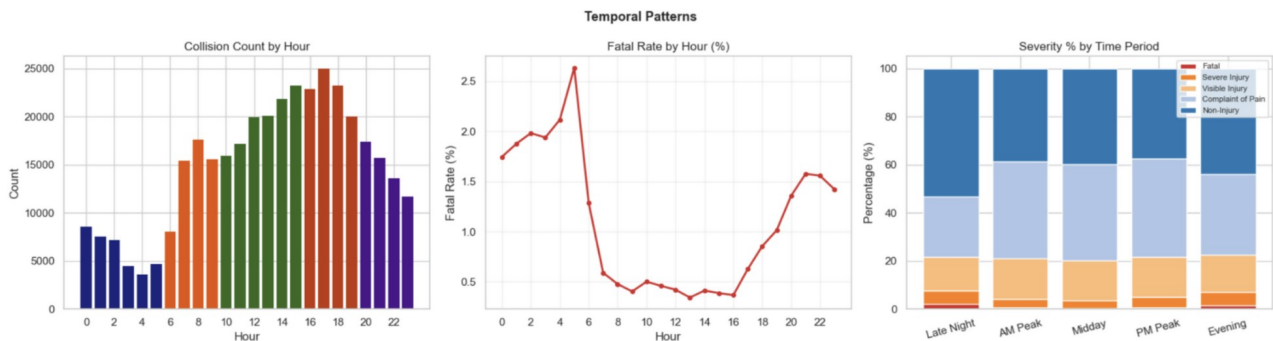


Figure 1. Temporal Patterns: Hourly Collision Count, Fatal Rate by Hour & Severity % by Time Period

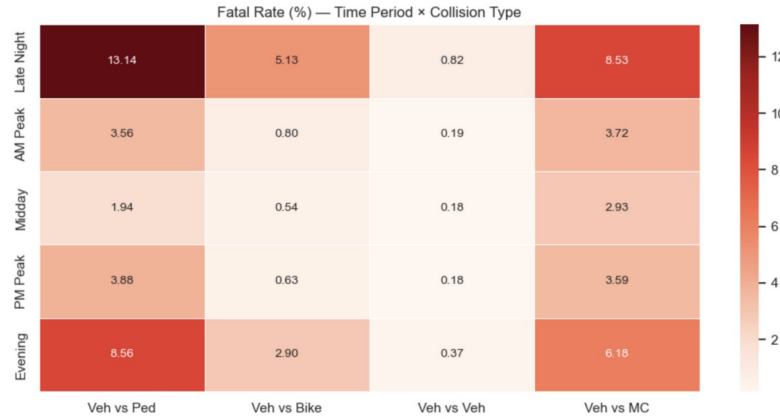


Figure 2. Fatal Rate (%) – Time Period × Collision Type

3.3 Demographic Analysis Results

The Fatal Rate varied by victim demographics. By age group, the pattern was U-shaped: victims under 18 and victims aged 65+ had the highest Fatal Rates, at approximately 1.78% and 1.82%, respectively, while middle-range-age groups (18–50) had the lowest, around 0.62% to 0.69%. Male victims had a higher Fatal Rate than female victims, 1.04% compared with 0.58%, approximately 1.7 times higher. By descent, Black victims had the highest Fatal Rate at approximately 1.03%, followed by Hispanic victims at 0.94%, while Asian victims had the lowest at approximately 0.74%. These differences may partly reflect spatial exposure and neighborhood-level traffic risk (see Figure 3).

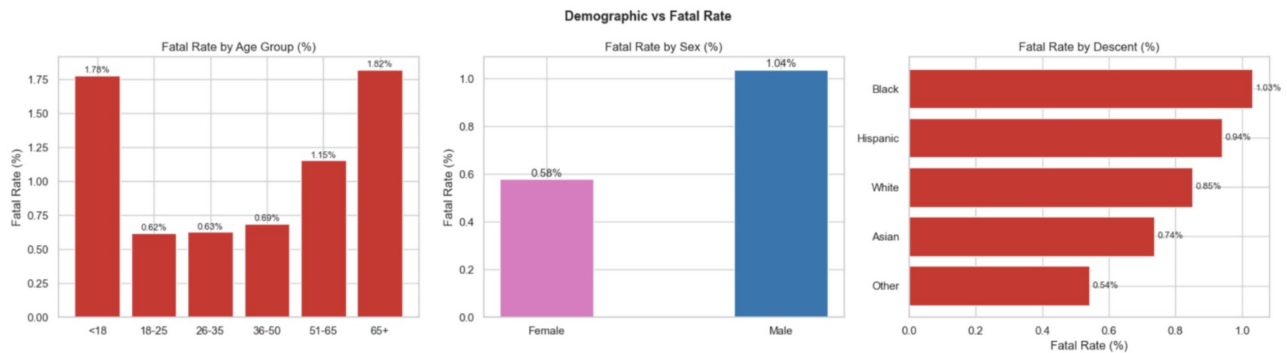


Figure 3. Demographic vs Fatal Rate: Fatal Rate by Age Group, Sex & Descent

3.4 Collision Type and MO Code Analysis Results

MO Code indicators revealed clear severity patterns. Vehicle-vs-pedestrian (Fatal Rate ~5.02%) and vehicle-vs-motorcycle (~4.07%) collisions far exceeded vehicle-vs-vehicle (~0.25%), confirming pedestrians and motorcyclists as the most vulnerable groups. Among contributing conditions, Street Racing showed the highest Fatal Rate (~34.44%), followed by DUI Felony (~3.57%) and Hit & Run (~1.55%) – a stark contrast to non-felony conditions.

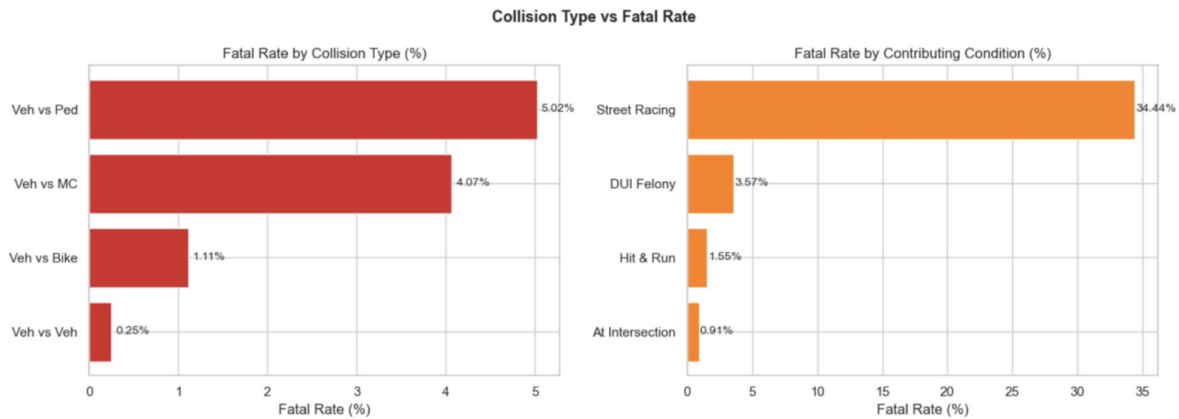


Figure 4. Collision Type & Contributing Conditions vs Severity: Fatal Rate by Type & Condition

Cramér's V, which measures association strength between categorical variables, identified vehicle-vs-pedestrian ($V = 0.33$) and Hit & Run Felony ($V = 0.34$) as the strongest predictors of collision severity among all MO Code features tested, both indicating moderate-strength associations.

3.5 Spatial Analysis Results

The KDE heat map shows that the overall collision hotspots are concentrated in the area from the city center to the southern part of Los Angeles, distributed along the main roads. However, the distribution of hotspots for fatal accidents and the overall collision density do not overlap – in eastern jurisdictions such as Hollenbeck and Newton, as well as port areas, the total number of accidents is not high, but fatal accidents are relatively concentrated (Figure 5). This indicates that simply using accident density to measure risk would underestimate the actual danger level in the eastern region. Therefore, we construct a spatial risk score based on the historical fatal rate per LAPD division, normalized to $[0, 1]$, as a model feature.

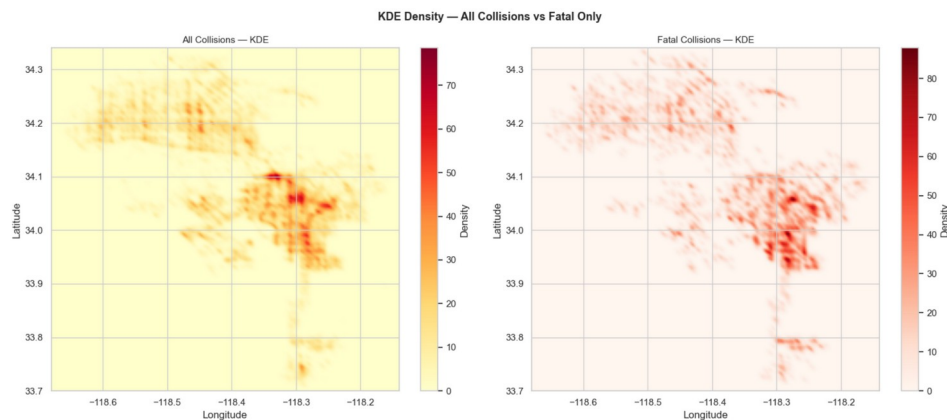


Figure 5. Collision Density KDE – All Collisions (left) vs Fatal Collisions only (right)

3.6 Random Forest Model Results

Table 1. Validation and Test Set Performance by Severity Class

Severity Class	Validation Precision	Validation Recall	Validation F1	Test Precision	Test Recall	Test F1
Fatal	0.1067	0.1830	0.1348	0.1195	0.2026	0.1504
Severe Injury	0.1991	0.3077	0.2418	0.2024	0.3119	0.2455
Visible Injury	0.3365	0.2860	0.3092	0.3450	0.3012	0.3216
Complaint of Pain	0.6443	0.4563	0.5342	0.6446	0.4570	0.5348
Non-Injury	0.6750	0.8489	0.7520	0.6776	0.8459	0.7524
Macro Avg	0.3923	0.4164	0.3944	0.3978	0.4237	0.4009
Accuracy	0.5816			0.5835		

The Random Forest model achieved 58.16% validation accuracy and 58.35% test accuracy, with macro F1-scores of 0.3944 and 0.4009, respectively. The similar validation and test results suggest stable performance with no strong signs of overfitting. Because of class imbalance, macro F1-score and per-class recall were more meaningful than accuracy alone.

For high-severity cases, the model achieved a Fatal recall of 0.2026, correctly identifying about 1 in 5 Fatal cases, and a Severe Injury recall of 0.3119, identifying about 1 in 3 Severe Injury cases. The model performed best on the Non-Injury class, with a recall of 0.8459 and an F1-score of 0.7524. Complaint of Pain and Visible Injury showed moderate performance, with F1-scores of 0.5348 and 0.3216. Fatal precision was low at 0.1195, meaning many predicted Fatal cases were actually lower-severity collisions, reflecting the common precision-recall tradeoff when modeling rare outcomes.

The confusion matrix in Figure 6 shows that the model often underestimates severity. Among actual Fatal cases, 93 were correctly classified, while 154 were misclassified as Severe Injury and 85 as Non-Injury.

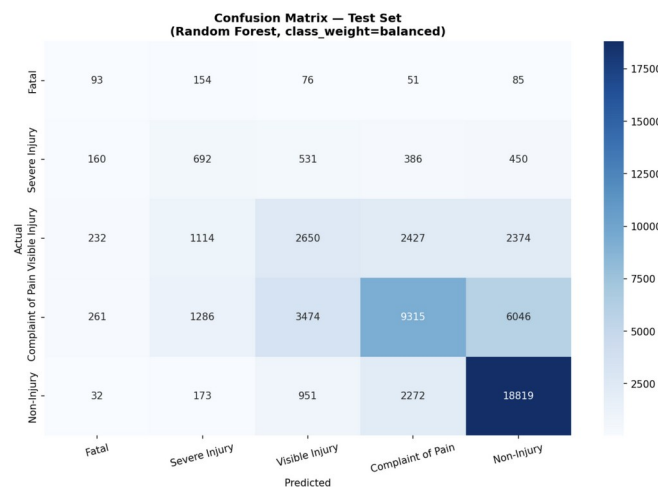


Figure 6. Confusion Matrix on Test Set

3.7 Young Female Sub-group Analysis

A subgroup analysis of young female victims aged 15 to 25 (with a total of 24,898 records) revealed that this group mainly suffered minor injuries - "pain complaints" accounted for 40.8%, "no injuries" accounted for 39.1%, and there were only 91 fatal cases (0.37%), which was lower than 0.85% of the overall dataset. The prediction results of the model are consistent with this: the average prediction probabilities of these two types are the highest (0.3152 and 0.3144, respectively), and the fatal probability is only 0.0310 (Figure 7). The model assigned approximately 63% of the probability to these two low-severity categories, which is consistent with the actual proportion of 79.9%. This indicates that the model has effectively captured the pattern of relatively mild injuries among young women, while still maintaining a certain ability to identify the risk of serious injuries.

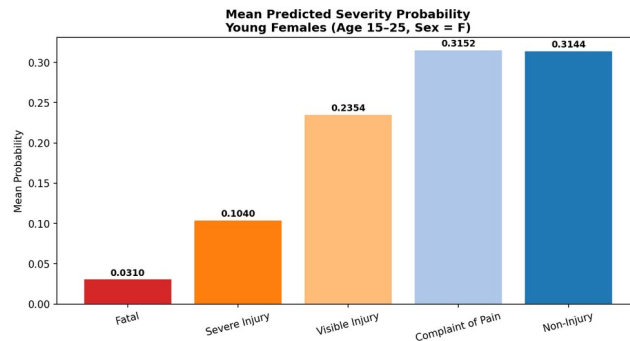


Figure 7. Mean Predicted Severity Probability for Young Female Victims (Age 15–25, Sex = F)

Section 4: Observation and Conclusion

The project found that the severity of traffic collisions in Los Angeles is jointly determined by factors such as time, space, population, and collision conditions. High-collision zones do not overlap with high-mortality zones: Newton, Hollenbeck, and Harbor have a lower total collision volume, but a higher mortality rate. The peak of accidents occurs in the afternoon, but the time with the highest mortality rate is at 5 a.m. Late-night vehicle-pedestrian collisions reached a Fatal Rate of 13.14%, while the overall Fatal Rate peaked at 2.55% at 5 a.m.. In collision situations, street racing (34.41%) and drunk driving felony (3.57%) are associated with the highest risk of death, and vehicle-to-pedestrian collisions (5.02%) are much higher than vehicle-to-vehicle collisions (0.25%).

The test accuracy of the random forest classifier is 58.35%, and its macroscopic F1 score is 0.4009, which is significantly higher than the majority class baseline of approximately 0.20, confirming that the engineered feature set carries the true prediction signal. Fatal recall of 0.2026 reflects not only class imbalance but also the inherent unpredictability of fatal outcomes. Absent

features such as vehicle speed and emergency response time likely determine the difference between Fatal and Severe Injury cases, suggesting that 0.2026 may approach the practical ceiling achievable with the current feature set. Notably, misclassified Fatal cases were most often assigned to Severe Injury rather than Non-Injury, indicating the model retains sensitivity to high-risk patterns even when exact classification fails.

The results suggest a qualified yes to the research question: the model demonstrated meaningful predictive ability overall (Macro F1: 0.4009) and successfully captured demographic-specific severity patterns, as shown in the young female subgroup analysis below. However, predicting rare high-severity outcomes such as Fatal cases remains challenging, reflecting limitations of the available feature set rather than the modeling approach itself.

The young female subgroup analysis (age 15-25, $n = 24,898$, Fatal Rate: 0.37% vs. overall 0.85%) confirmed that demographic features function as genuine predictive signals. The model assigned 63% of predicted probability to the two lowest severity classes, closely matching the actual distribution of 79.9%, directly supporting the claim that demographic characteristics contribute meaningfully to severity prediction.

Overall, this project indicates that the number of collisions alone cannot represent traffic safety risks. Future work should prioritize improving the fatal recall rate through SMOTE oversampling or re-establishment of binary classification, and expand subgroup analyses across population groups to further verify population sensitivity.

Section 5: References

- Alam, M. S., & Tabassum, N. J. (2023). Spatial pattern identification and crash severity analysis of road traffic crash hotspots. *Heliyon*, 9(5).
- Alsahfi, T. (2024). Spatial and temporal analysis of road traffic accidents in major Californian cities using a geographic information system. *ISPRS International Journal of Geo-Information*, 13(5), 157.
- Los Angeles Police Department. (2026). Traffic collision data from 2010 to present [Data set]. *Los Angeles Open Data Portal*. Retrieved May 6, 2026, from https://data.lacity.org/Public-Safety/Traffic-Collision-Data-from-2010-to-Present/d5tf-ez2w/about_data